Complete FYP Idea Discovery, Proposal Defense, and Project Execution Prompt Workflow

Student Guide for a Two-Semester Final Year Project

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1. Purpose of This Guide

This guide provides a complete, multi-step workflow that student groups can use with an AI system to:

- discover possible FYP problems;
- investigate stakeholders;
- validate whether a problem is real;
- study contemporary systems;
- investigate datasets;
- compare ideas with previous university FYPs;
- test technical feasibility;
- reduce project scope;
- design evaluation plans;
- prepare for the August proposal defense;
- plan two FYP-I development iterations;
- complete the practical system during FYP-II;
- prepare for the university job fair;
- and finish formal testing, deployment, and documentation.

The workflow is designed for AI systems with:

- file-reading capabilities;
- online search;
- access to current research;
- access to product and technology information;
- and the ability to compare and evaluate multiple alternatives.

The workflow should not be used to generate a quick list of attractive project titles.

Its purpose is to identify a **viable, defensible, service-oriented FYP** that can survive proposal evaluation and be completed within the university timeline.

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2. Assumptions About the FYP Process

This workflow assumes the following:

- Student groups have already been formed.
- All groups are currently trying to develop an FYP idea.
- The proposal defense takes place in August.
- The August defense determines whether the group has a viable FYP idea.
- Students must investigate the idea before the August defense.
- After approval, students begin project implementation.
- FYP-I should contain two development iterations.
- By the end of FYP-I, students should have a demonstrable prototype.
- FYP-II should bring the project to practical completion by mid-April.
- The university job fair takes place around mid-April.
- Companies visit student booths during the job fair.
- The period from mid-April to May is mainly used for:
- formal testing;
- deployment;
- documentation;
- report completion;
- final corrections;
- and final university evaluation.

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3. Correct FYP Lifecycle

Phase A: Pre-August Proposal Development

Before the proposal defense, students should complete:

- domain investigation;
- stakeholder identification;
- current-workflow analysis;
- problem validation;
- contemporary-system comparison;
- research comparison;
- dataset investigation;
- previous-FYP comparison;
- technical feasibility analysis;
- scope definition;
- initial architecture;
- risk analysis;
- and evaluation planning.

The objective of this phase is not to build the project.

The objective is to determine whether the proposed project is sufficiently:

- useful;
- differentiated;
- technically meaningful;
- feasible;
- measurable;
- and appropriately scoped.

Phase B: August Proposal Defense

During the defense, the group should be prepared to explain:

1. Which domain the project belongs to
2. Who the stakeholders are
3. What problem the stakeholders experience
4. How the current workflow operates
5. What evidence shows that the problem exists
6. Which contemporary systems already address it
7. Why those systems are insufficient
8. Which previous university FYPs are similar
9. How the proposed project differs
10. Which datasets are available
11. Whether the data can legally and practically be used
12. What the main technical contribution will be
13. What service the proposed system will provide
14. How the project will be evaluated
15. What will be completed in FYP-I
16. What will be completed in FYP-II
17. Why the project can be completed within the available time

The group should not enter the defense with only:

- a title;
- a general domain;
- a feature list;
- or an intention to “use AI.”

Phase C: FYP-I

After proposal approval, implementation begins.

FYP-I should contain two development iterations.

FYP-I Iteration 1

The purpose of Iteration 1 is to answer:

> Can the central technical and service idea work?

It should normally include:

- initial data acquisition;
- data inspection;
- a baseline method;
- a proof of concept;
- basic backend or processing logic;
- an initial user interface;
- and one limited end-to-end workflow.

FYP-I Iteration 2

The purpose of Iteration 2 is to answer:

> Can the idea work as an integrated prototype?

It should normally include:

- an improved technical component;
- integration of the main modules;
- a basic database;
- a usable interface;
- initial testing;
- user or stakeholder feedback where possible;
- and a demonstrable service workflow.

End of FYP-I expectation

By the end of FYP-I, the group should have:

- completed two iterations;
- validated technical feasibility;
- prepared or acquired important data;
- implemented major parts of the architecture;
- and produced a demonstrable prototype.

The FYP-I demonstration should not consist only of:

- mock-ups;
- diagrams;
- static screens;
- disconnected notebooks;

- or an unintegrated model.

Phase D: FYP-II

FYP-II should convert the prototype into a complete, evaluated, and reliable system.

The group should:

- complete remaining core functions;
- improve the technical component;
- integrate all essential modules;
- complete the main service workflow;
- perform technical evaluation;
- conduct user or stakeholder testing;
- improve reliability;
- and prepare a stable demonstration.

The practical system should be substantially complete by mid-April.

Phase E: Mid-April Job Fair

By the university job fair, the project should have:

- a stable working system;
- a clear problem statement;
- a complete main workflow;
- a visible technical contribution;
- meaningful evaluation results;
- a professional interface;
- and a reliable demonstration.

Students should be able to repeatedly demonstrate the system to visiting companies.

No essential project feature should remain dependent on future development.

Phase F: Mid-April to May

This period should be used for:

Formal testing

- functional testing;
- integration testing;

- regression testing;
- performance testing;
- security testing;
- usability testing;
- failure-case testing;
- and final evaluation improvements.

Deployment

- server or cloud deployment;
- local deployment where required;
- database migration;
- model deployment;
- API configuration;
- backup procedures;
- monitoring;
- logging;
- and environment documentation.

Documentation

- final report;
- architecture documentation;
- methodology;
- experiments;
- evaluation results;
- limitations;
- user manuals;
- installation instructions;
- deployment instructions;
- and final presentation material.

Final refinement

- corrections based on job-fair feedback;
- reliability improvements;
- interface polish;
- deployment fixes;
- and final submission preparation.

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4. How Students Should Use This Workflow

The workflow should be executed one stage at a time.

Do not paste every prompt into the AI simultaneously.

For each stage:

1. Read the purpose of the stage.
2. Prepare the required inputs.
3. Copy the prompt.
4. Attach relevant files.
5. provide outputs from earlier stages;
6. inspect the response;
7. correct incorrect assumptions;
8. save the accepted output;
9. and then move to the next stage.

A group should maintain a shared workflow folder.

Suggested structure:

```
FYP-Workflow/
│
├── Phase-A-Pre-Defense/
│   ├── 01-Repository-Intelligence/
│   ├── 02-Group-Capability-Profile/
│   ├── 03-Domain-Exploration/
│   ├── 04-Problem-Opportunities/
│   ├── 05-Stakeholder-Validation/
│   ├── 06-Problem-Evidence/
│   ├── 07-Contemporary-Systems/
│   ├── 08-Dataset-Investigation/
│   ├── 09-Opportunity-Ranking/
│   ├── 10-Solution-Directions/
│   ├── 11-Service-Concepts/
│   ├── 12-Novelty-Audit/
│   ├── 13-Technical-Feasibility/
│   ├── 14-Scope-Evaluation-Risk/
│   └── 15-Proposal-Defense-Package/
│
├── Phase-B-Post-Defense/
│   └── 16-Defense-Feedback-Revision/
│
├── Phase-C-FYP-I/
│   ├── 17-Iteration-1/
│   ├── 18-Iteration-2/
│   └── 19-FYP-I-Prototype-Review/
│
└── Phase-D-FYP-II/
    ├── 20-FYP-II-Completion/
    ├── 21-Evaluation/
```

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    ├── 22-Job-Fair-Readiness/
    └── 23-Testing-Deployment-Documentation/
```

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5. Information Required Before Starting

Input A: Previous FYP Repository

Attach the university repository containing previous FYPs.

For example:

```
BS-FYP-Repo-Fall2019-Fall2025.md
```

This is used to:

- understand university expectations;
- identify repeated concepts;
- find saturated domains;
- study recent FYP trends;
- compare proposed ideas with previous projects;
- and test whether a new concept is meaningfully different.

Input B: Group Capability Profile

The group is already formed.

Complete the following:

```
Number of group members:

Degree programme:

Member 1 skills:

Member 2 skills:

Member 3 skills:

Other members, if applicable:
```

```
Strongest programming languages:

Web-development experience:

Mobile-development experience:

Machine-learning experience:

Data-science experience:

Cybersecurity experience:

Cloud-computing experience:

IoT or hardware experience:

Game-development experience:

UI/UX experience:

Database experience:

Software-testing experience:

Available laptops:

Available GPU and GPU memory:

Available cloud services:

Maximum cloud or API budget:

Maximum hardware budget:

Access to organizations:

Access to domain experts:

Access to potential users:

Preferred project domains:

Domains the group wants to avoid:

Preferred project type:
- Development
- Research
- Hybrid
```

```
Willingness to collect data:

Willingness to label data:

Willingness to conduct interviews:

Willingness to perform user studies:

Technologies the group wants to learn:

Technologies the group wants to avoid:

Expected distribution of responsibilities:

Additional limitations:
```

Students should answer honestly.

The purpose is not to make the group appear highly skilled.

The purpose is to prevent the selection of an unrealistic project.

Input C: University and Supervisor Requirements

Complete the following where possible:

```
Proposal-defense date:

Expected proposal-defense content:

Expected number of modules:

Expected technical complexity:

Preferred project methodology:

FYP-I duration:

Expected number of FYP-I iterations:
Two

FYP-I evaluation requirements:
```

```
Expected FYP-I prototype:

FYP-II duration:

Job-fair date:

Final evaluation date:

Documentation requirements:

Deployment expectations:

Testing expectations:

Research requirement:

Publication expectation:

Ethics-review requirements:

Supervisor’s preferred domains:

Supervisor’s preferred technologies:

Supervisor’s restrictions:

Other departmental requirements:
```

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6. Rules That Apply to Every Stage

6.1 Problem-first rule

Do not start by selecting a technology.

Avoid beginning with:

- “What can we build using an LLM?”
- “Where can blockchain be used?”
- “What project can use computer vision?”
- “How can we add an AI agent?”
- “What can we make using RAG?”

Begin with:

- Who is experiencing a problem?
- What are they trying to accomplish?
- How is the task currently performed?
- Where does the workflow fail?
- What are the consequences?
- Why are existing solutions insufficient?

Technology should be selected only after the problem is understood.

6.2 Service-oriented rule

A strong FYP should normally provide an end-to-end service.

A complete service usually contains:

1. A trigger or user need
2. Data or user input
3. Validation
4. Technically meaningful processing
5. An actionable output
6. A person or organization that acts on the output
7. Follow-up or status tracking
8. Outcome measurement
9. Reporting or learning

Weak example:

> A model that detects potholes.

Stronger example:

> A road-maintenance service that receives images and locations, detects probable road damage, estimates severity, combines duplicate reports, prioritizes maintenance work, provides an authority dashboard, assigns cases, and tracks resolution.

6.3 Evidence rule

Online search should be used to verify:

- whether a problem exists;
- how serious it is;
- which stakeholders are affected;

- contemporary systems;
- recent research;
- open-source alternatives;
- available datasets;
- APIs;
- hardware requirements;
- laws;
- regulations;
- costs;
- and evaluation methods.

Prefer:

- official government sources;
- official institutional sources;
- peer-reviewed research;
- conference papers;
- official technical documentation;
- official dataset repositories;
- recognized industry reports;
- and credible journalism.

Avoid relying mainly on:

- random blogs;
- AI-generated idea websites;
- anonymous posts;
- promotional articles;
- and unsupported social-media claims.

Every important online claim should include:

- source;
- publication date;
- and access date where appropriate.

6.4 Novelty rule

Do not call an idea novel merely because its exact title is absent.

Novelty should be investigated through:

- previous university FYPs;
- contemporary commercial systems;
- open-source projects;
- research papers;

- datasets;
- and actual stakeholder needs.

The following do not automatically create novelty:

- changing from web to mobile;
- changing the city;
- translating an interface;
- adding Urdu support;
- using a newer model;
- adding a chatbot;
- adding RAG;
- adding blockchain;
- adding an AI agent;
- or combining several familiar features.

6.5 Feasibility rule

A project should not be recommended unless:

- the data can be obtained;
- the main stakeholders can be reached or realistically simulated;
- the project can survive the August defense;
- a technical proof of concept can be developed during FYP-I;
- two FYP-I iterations can be completed;
- a demonstrable prototype can be produced by the end of FYP-I;
- the practical system can be completed by mid-April;
- and testing, deployment, and documentation can be completed by May.

6.6 Common ideas that require strong differentiation

The following topics are common and should not be accepted without substantial improvement:

- generic chatbots;
- basic RAG systems;
- document question-answering tools;
- generic recommendation systems;
- ordinary marketplaces;
- attendance systems;
- learning-management systems;
- simple disease classifiers;
- plant-disease detection;
- fake-news detection;
- deepfake detection;

- generic career assistants;
- generic university assistants;
- basic mental-health chatbots;
- generic legal chatbots;
- standard e-commerce applications;
- ordinary productivity assistants;
- and basic content-generation tools.

Phase A: Pre-August Idea Investigation and Proposal Preparation

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Stage 1: Repository Intelligence

Purpose

Understand what previous students built and what the university appears to consider an acceptable FYP.

Inputs

- Previous FYP repository

Prompt

```
You are acting as an FYP Repository Analyst.

Analyze the attached repository of previous university FYPs.

Do not generate new FYP ideas during this stage.

First normalize the repository:

1. Identify unique projects.
2. Detect repeated student-member rows.
3. Detect duplicate tables.
4. Detect repeated project entries.
5. Group projects by year and cohort.
6. Separate development, research, and hybrid projects where possible.
7. Identify missing or inconsistent information.
8. Explain how repository inconsistencies affect the analysis.
```

```
Create a Repository Intelligence Profile containing:

A. Dominant problem domains
B. Common stakeholders
C. Common service architectures
D. Common technical approaches
E. Frequently repeated ideas
F. Saturated project areas
G. Underrepresented problem spaces
H. Changes across different years
I. Detailed patterns in the most recent two years
J. Common problem-formulation approaches
K. Common project-description weaknesses
L. What appears to make a project acceptable as an FYP
M. Differences between stronger and weaker projects
N. Development-project patterns
O. Research-project patterns
P. Common scope problems
Q. Common data-access risks
R. Common evaluation weaknesses
S. Common deployment weaknesses
T. Common job-fair demonstration strengths and weaknesses where inferable

For every major conclusion, provide examples from the repository.

Finish with:

1. Topics that should normally be avoided
2. Topics that require substantial differentiation
3. Potentially underexplored problem spaces
4. Characteristics a new FYP should contain
5. A framework for comparing a proposed idea with earlier FYPs
6. Questions students should expect during the August proposal defense

Do not call an idea novel merely because its exact title does not occur.
```

Output

Save as:

`Repository Intelligence Profile`

Gate

Do not proceed until the repository has been analyzed and saturated topics have been identified.

Stage 2: Group Capability and Constraint Profile

Purpose

Determine what the already-formed group can realistically execute.

Inputs

- Group Capability Profile
- University requirements
- Repository Intelligence Profile

Prompt

```
You are acting as an FYP Group Feasibility Adviser.

The group has already been formed.

Review:

1. The skills of each group member
2. Available hardware
3. Available GPU memory
4. Budget
5. Access to cloud services
6. Access to organizations
7. Access to domain experts
8. Access to potential users
9. University requirements
10. The August proposal defense
11. The two required FYP-I iterations
12. The required FYP-I prototype
13. The mid-April practical completion target
14. The May testing, deployment, and documentation deadline

Do not generate project ideas yet.

Create a Group Capability and Constraint Profile containing:

1. Existing strengths
2. Skill gaps
3. Skills that can realistically be learned
4. Technologies that create excessive risk
5. Realistic model-training capacity
6. Realistic inference capacity
```

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7. Realistic web-development capacity
8. Realistic mobile-development capacity
9. Realistic data-engineering capacity
10. Realistic testing and deployment capacity
11. Data-collection capacity
12. Access to stakeholders
13. Access to experts
14. Financial limitations
15. Privacy, legal, and ethical limitations
16. Suitable project domains
17. Unsuitable project domains
18. Suitable project types:
    - Development
    - Research
    - Hybrid
19. Recommended project complexity
20. Recommended maximum number of major technical components
21. Recommended division of responsibilities
22. Risks specific to the group
23. Assumptions that still require verification

Classify potential project types as:

A. Safely achievable
B. Achievable with controlled risk
C. High risk
D. Unsuitable

End with non-negotiable constraints that all later stages must obey.

The group must be able to:

- defend the idea in August;
- complete two meaningful FYP-I iterations;
- demonstrate a prototype at the end of FYP-I;
- complete the practical system by mid-April;
- and complete formal testing, deployment, and documentation by May.
```

Output

`Group Capability and Constraint Profile`

Stage 3: Domain Exploration

Purpose

Identify promising domains before jumping to specific applications.

Inputs

- Repository Intelligence Profile
- Group Capability and Constraint Profile
- Online search

Prompt

```
You are acting as an FYP Domain Exploration Researcher.

Using:

1. The Repository Intelligence Profile
2. The Group Capability and Constraint Profile
3. Current online evidence
4. Pakistani and local contexts where relevant

Identify 12 to 20 possible domains that may contain suitable FYP problems.

Possible domains may include:

- education;
- healthcare;
- agriculture;
- climate and environment;
- transportation;
- cybersecurity;
- law and justice;
- public services;
- accessibility;
- small businesses;
- finance;
- manufacturing;
- supply chains;
- energy;
- disaster response;
- workplace operations;
- university operations;
- community services;
```

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- media;
- and human-computer interaction.

For each domain, provide:

1. Domain name
2. Main stakeholder groups
3. Important current workflows
4. Common difficulties
5. Local or Pakistani relevance
6. Recent technological changes
7. Existing products or platforms
8. Previous FYP concentration
9. Data-access potential
10. Stakeholder-access potential
11. Technical-depth potential
12. Suitability for this group
13. Main risk
14. Why the domain should or should not proceed

Do not generate specific FYP ideas yet.

Rank the domains and retain 6 to 10 for deeper investigation.
```

Output

`Domain Opportunity Map`

Stage 4: Problem-Space Discovery

Purpose

Discover genuine stakeholder problems rather than technology-led ideas.

Inputs

- Domain Opportunity Map
- Repository Intelligence Profile
- Group Capability Profile
- Online search

Prompt

```
You are acting as a Problem-Space Researcher.

Use the selected domains to identify real workflows in which a clearly defined
stakeholder experiences:

- delays;
- repetitive work;
- fragmented information;
- high cost;
- poor accessibility;
- weak coordination;
- lack of personalization;
- delayed detection;
- preventable errors;
- poor transparency;
- limited expert access;
- privacy risks;
- information overload;
- poor monitoring;
- weak decision support;
- inefficient resource use;
- or difficult reporting.

Generate 25 to 40 problem opportunities.

Do not propose solutions yet.

For every problem opportunity, provide:

1. Problem name
2. Domain
3. Primary stakeholder
4. Secondary stakeholders
5. Current workflow
6. Exact point of failure
7. Consequences
8. Existing workaround
9. Frequency
10. Scale
11. Local relevance
12. Evidence that the problem exists
13. Possible stakeholder access
14. Possible data availability
15. Similar previous FYPs
```

```
16. Why it may suit a computing FYP
17. Main uncertainty
18. Main reason it may fail proposal defense

Bad problem formulation:

“There is no AI chatbot for farmers.”

Better problem formulation:

“Small farmers receive delayed and generic advice because agricultural experts
cannot inspect every field or respond to every case individually.”

Do not attach a specific model, application type, or fashionable technology to
the problem.
```

Output

`Problem Opportunity Catalogue`

Stage 5: Stakeholder and Workflow Validation

Purpose

Ensure that the proposed problem is connected to real users and a real workflow.

Inputs

- Problem Opportunity Catalogue
- Online search
- Interviews or informal stakeholder discussions where possible

Prompt

```
You are acting as a Stakeholder and Workflow Validation Researcher.

Review each problem opportunity.

For every problem, identify:

1. Primary stakeholder
2. Secondary stakeholders
3. Decision-makers
```

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4. Administrators or professionals
5. People affected indirectly
6. Current workflow
7. Inputs used in the workflow
8. People responsible for each step
9. Tools currently used
10. Delays
11. Errors
12. Costs
13. Communication failures
14. Data handoffs
15. Existing checks
16. Existing workarounds
17. Outcomes
18. Current performance indicators
19. Stakeholders the students can realistically contact
20. Stakeholders the students cannot realistically contact

Create a stakeholder-access verdict:

- Direct access available
- Indirect access available
- Simulated access possible
- Access uncertain
- Access unavailable

Create a workflow-validity verdict:

- Clearly understood
- Partially understood
- Poorly understood
- Based mainly on assumptions

Reject or downgrade problems where:

- the primary stakeholder is unclear;
- the workflow cannot be explained;
- no users can be accessed;
- or the problem is based mainly on assumptions.

Retain only problems with a defensible stakeholder and workflow.
```

Output

`Validated Stakeholder and Workflow Report`

Stage 6: Problem Evidence Validation

Purpose

Determine whether the problem is real, significant, and current.

Inputs

- Validated problems
- Online search
- Stakeholder evidence

Prompt

```
You are acting as a Problem Evidence Validation Researcher.

For every surviving problem, use reliable online sources and stakeholder
evidence to investigate:

1. Whether the problem currently exists
2. How many people or organizations may be affected
3. How frequently it occurs
4. Whether credible institutions recognize it
5. Economic, social, operational, or technical consequences
6. Current workarounds
7. Why current workarounds remain insufficient
8. Whether the problem is local, international, or both
9. Whether the problem is changing due to technology
10. Whether the problem is changing due to regulation
11. Whether a computing intervention is appropriate
12. Whether improvement can be measured
13. Whether students can study the problem before the August defense
14. Whether a practical prototype can be built in FYP-I
15. Whether the system can be completed by mid-April

Classify the evidence as:

- Strong
- Moderate
- Weak
- Contradictory
- Insufficient

For every problem, cite:
```

```
- official reports;
- research;
- credible institutional sources;
- or stakeholder evidence.

Reject problems that:

- rely mainly on assumptions;
- have no reliable evidence;
- are too broad;
- require inaccessible confidential information;
- cannot produce measurable improvement;
- or cannot reasonably become a computing FYP.

Produce a shortlist of 12 to 20 validated problems.
```

Output

`Validated Problem Shortlist`

Stage 7: Contemporary Systems and Research Investigation

Purpose

Understand what already exists before proposing a solution.

Inputs

- Validated Problem Shortlist
- Online search

Prompt

```
You are acting as a Contemporary Systems and Research Analyst.

For every validated problem, investigate:

A. Commercial systems

1. Current products
2. Target users
```

```
3. Main features
4. Pricing
5. Deployment model
6. Strengths
7. Weaknesses
8. Local availability
9. Local suitability
10. Data requirements

B. Institutional systems

1. Government platforms
2. University systems
3. Hospital systems
4. Industry systems
5. Non-profit systems
6. Local organizational tools

C. Open-source systems

1. Relevant repositories
2. Main capabilities
3. License
4. Technical requirements
5. Activity and maintenance
6. Limitations

D. Recent research

1. Relevant papers from approximately the last five years
2. Proposed methods
3. Datasets
4. Evaluation metrics
5. Limitations
6. Open research questions
7. Implementation complexity

For every problem, provide:

1. Closest commercial solution
2. Closest institutional solution
3. Closest open-source solution
4. Closest research direction
5. What is already solved
6. What remains poorly solved
7. What is not locally adapted
8. What is technically feasible for students
9. What should not be duplicated
```

```
10. Possible meaningful gaps

Do not propose novelty merely by removing a subscription fee or changing the
interface.
```

Output

`Contemporary Systems and Research Report`

Stage 8: Dataset and Data-Source Investigation

Purpose

Determine whether each possible project has realistic data.

Inputs

- Validated problems
- Contemporary Systems Report
- Online search

Prompt

```
You are acting as an FYP Dataset and Data-Source Investigator.

For every validated problem, investigate all required data.

For each potential dataset or data source, provide:

1. Dataset name
2. Dataset owner
3. Download or access method
4. License
5. Data format
6. Number of records
7. Number of classes
8. Date range
9. Geographic coverage
10. Language coverage
11. Local relevance
12. Data quality
13. Missing values
14. Class imbalance
```

```
15. Labels
16. Annotation quality
17. Privacy concerns
18. Ethical concerns
19. Commercial-use restrictions
20. Academic-use restrictions
21. Expected preprocessing
22. Expected storage requirements
23. Expected compute requirements
24. Whether the data support the proposed problem
25. Known limitations

Also investigate:

- APIs;
- sensor data;
- public records;
- documents;
- user-generated data;
- institutional data;
- synthetic data;
- and data the students may collect themselves.

For each problem, create:

A. Preferred data plan
B. Minimum viable data plan
C. Local-data plan
D. Fallback data plan
E. Data-acquisition deadline
F. Data risk rating

Reject or heavily penalize problems for which:

- no suitable data exist;
- data access depends on uncertain approval;
- licensing is unsuitable;
- annotation is impossible;
- or meaningful data cannot be prepared during FYP-I.

Do not recommend “we will collect the data later” without a specific collection
and annotation plan.
```

Output

`Dataset Investigation Report`

Stage 9: Opportunity Ranking

Purpose

Select the strongest problems before designing solutions.

Inputs

- Problem evidence
- Stakeholder report
- Contemporary systems report
- Dataset report
- Group capability profile

Prompt

```
You are acting as an FYP Opportunity Evaluation Committee.

Score each validated problem from 1 to 10.

Use the following weights:

1. Problem importance – 12%
2. Frequency or scale – 8%
3. Stakeholder clarity – 8%
4. Stakeholder access – 8%
5. Local relevance – 5%
6. Evidence strength – 10%
7. Data feasibility – 12%
8. Technical-depth potential – 10%
9. End-to-end service potential – 8%
10. Evaluation potential – 8%
11. Differentiation potential – 6%
12. Group capability match – 5%

Apply penalties for:

- inaccessible users;
- inaccessible institutions;
- unavailable data;
- unclear workflows;
- high privacy risk;
- paid-API dependence;
- excessive hardware requirements;
```

```
- weak local relevance where local adaptation is claimed;
- similarity to previous FYPs;
- and inability to produce a prototype during FYP-I.

For every score, provide a justification.

Select the top 8 to 12 problems.

For each retained problem, explain:

1. Why it survived
2. Strongest FYP potential
3. Main feasibility risk
4. Main novelty risk
5. Main data risk
6. Main proposal-defense question
7. Evidence still required
```

Output

`Ranked FYP Problem Opportunities`

Stage 10: Solution-Direction Generation

Purpose

Generate several fundamentally different ways to address each problem.

Inputs

- Ranked problems
- Contemporary systems report
- Dataset report
- Group constraints

Prompt

```
You are acting as a Solution-Direction Generator.

For each selected problem, generate four to six fundamentally different
computing interventions.

Do not create minor variations of the same application.
```

```
Possible intervention types include:

- workflow automation;
- decision support;
- monitoring and alerts;
- intelligent retrieval;
- forecasting;
- optimization;
- recommendation;
- matching;
- expert triage;
- simulation;
- cybersecurity controls;
- data-quality improvement;
- human-in-the-loop assistance;
- accessibility technology;
- multimodal analysis;
- privacy-preserving systems;
- intelligent resource allocation;
- explainable decision support;
- or agent-assisted workflow execution.

For every solution direction, explain:

1. Core intervention
2. Why it addresses the underlying problem
3. Primary user
4. Secondary users
5. Required data
6. Main technical contribution
7. Service output
8. Human role
9. Proposed workflow
10. Potential evaluation
11. Main risk
12. Main dependency
13. Difference from other directions
14. Similar existing systems
15. Similar previous FYPs
16. Whether the central idea can be prototyped in FYP-I
17. Whether the system can be completed by mid-April

Explicitly answer:

- Would the system still be useful without AI?
- What does AI or the technical component genuinely improve?
- Is the proposed technology necessary?
```

```
- Is the direction only a feature?
- Can it become an end-to-end service?
- Is it sufficiently differentiated?
- Can it survive the August proposal defense?

Select the two strongest directions for each problem.
```

Output

`Solution Direction Portfolio`

Stage 11: Service-System Design

Purpose

Convert promising technical directions into complete FYP services.

Inputs

- Solution directions
- Stakeholder workflows
- Dataset report
- Group constraints

Prompt

```
You are acting as an FYP Service-System Architect.

Convert every surviving solution direction into an end-to-end service concept.

For each concept, define:

1. Working title
2. One-sentence concept
3. Domain
4. Problem statement
5. Primary stakeholder
6. Secondary stakeholders
7. Existing workflow
8. Proposed workflow
9. Trigger that starts the service
10. User inputs
11. Data sources
```

```
12. Data validation
13. Main technical component
14. Output produced
15. Person or organization acting on the output
16. Human review
17. Follow-up
18. Outcome tracking
19. Records maintained
20. Administrative oversight
21. Feedback mechanism
22. Main user interface
23. Administrative interface
24. External integrations
25. Privacy requirements
26. Security requirements
27. Core value proposition
28. Expected operational improvement
29. Minimum FYP-I proof of concept
30. End-of-FYP-I prototype
31. Mid-April complete system
32. Post-job-fair testing and deployment work

Represent the workflow as:

Trigger
→ Data collection
→ Validation
→ Technical processing
→ Recommendation, alert, or action
→ Human review where required
→ Follow-up
→ Outcome tracking
→ Reporting or learning

Reject concepts that remain only:

- classifiers;
- chatbots;
- RAG interfaces;
- dashboards;
- content generators;
- API wrappers;
- or disconnected feature collections.

Every concept must solve a complete workflow problem.
```

Output

`FYP Service Concept Portfolio`

Stage 12: Novelty and Duplication Audit

Purpose

Compare each concept against previous FYPs and external systems.

Inputs

- Service concepts
- Previous FYP repository
- Contemporary systems report
- Online search

Prompt

```
You are acting as an FYP Novelty and Duplication Auditor.

For every service concept, conduct four comparisons:

A. Previous university FYP comparison
B. Commercial-system comparison
C. Open-source-system comparison
D. Recent research comparison

For the university repository:

1. Identify the five most similar FYPs.
2. Compare stakeholders.
3. Compare problem statements.
4. Compare current workflows.
5. Compare proposed workflows.
6. Compare features.
7. Compare technical methods.
8. Compare datasets.
9. Compare evaluation methods.
10. Compare expected outputs.

For external systems:
```

```
1. Identify the closest commercial product.
2. Identify the closest institutional platform.
3. Identify the closest open-source project.
4. Identify the closest research direction.
5. Identify existing datasets and benchmarks.
6. Identify limitations that remain unsolved.

Classify each concept as:

- Highly repetitive
- Mostly repetitive
- Moderately differentiated
- Strongly differentiated
- Potentially novel but unverified

Do not accept the following as sufficient novelty:

- changing the city;
- changing web to mobile;
- translating the application;
- adding a chatbot;
- adding RAG;
- adding an agent;
- adding blockchain;
- using a newer model;
- adding more features;
- or combining several existing tools without a coherent contribution.

For each concept, state:

1. Closest previous FYP
2. Closest existing product
3. Closest research direction
4. Genuine differentiating contribution
5. Whether the difference matters to stakeholders
6. Whether the contribution is:
   - technical;
   - operational;
   - local;
   - data-based;
   - usability-based;
   - or research-based
7. Evidence against novelty
8. Proposal-defense risk
9. Final novelty verdict

Reject highly repetitive concepts.
```

Output

`Novelty and Duplication Audit`

Stage 13: Technical Feasibility and Initial Architecture

Purpose

Determine whether the concept can actually be implemented.

Inputs

- Service concepts
- Dataset report
- Group capability profile
- Novelty audit
- Online search

Prompt

```
You are acting as a Technical Feasibility and System Architecture Analyst.

For every surviving concept, investigate:

A. Data feasibility

1. Required datasets
2. Public datasets
3. Local-data requirements
4. Data licenses
5. Data size
6. Annotation requirements
7. Collection method
8. Privacy issues
9. Class imbalance
10. Minimum viable dataset
11. Data-acquisition deadline
12. Fallback plan

B. Model or algorithm feasibility
```

```
1. Candidate approaches
2. Baseline methods
3. Pretrained models
4. Fine-tuning needs
5. Training requirements
6. Inference requirements
7. GPU-memory requirements
8. CPU alternatives
9. Offline alternatives
10. Expected limitations
11. Explainability requirements
12. Failure risks
13. Hallucination risks where relevant

C. System feasibility

1. Frontend
2. Backend
3. Database
4. Data pipeline
5. Intelligent component
6. Authentication
7. Authorization
8. APIs
9. Deployment
10. Logging
11. Monitoring
12. Backup
13. Security
14. Integration risks

D. Operational feasibility

1. Stakeholder access
2. Expert access
3. User-testing access
4. Required permissions
5. Institutional dependencies
6. Legal issues
7. Ethical issues

E. Financial feasibility

1. API costs
2. Cloud costs
3. Hardware costs
4. Licensing costs
5. Data costs
```

```
6. Expected total budget

F. FYP timeline feasibility

Determine whether the group can:

1. Defend the concept in August
2. Complete Iteration 1 during FYP-I
3. Complete Iteration 2 during FYP-I
4. Demonstrate a prototype at the end of FYP-I
5. Complete the practical system by mid-April
6. Complete testing, deployment, and documentation by May

Create an initial architecture containing:

- user roles;
- frontend;
- backend;
- database;
- intelligent component;
- data flow;
- external services;
- deployment environment;
- and security boundaries.

Produce:

1. Feasibility verdict
2. Critical dependencies
3. Most likely failure point
4. Lower-risk alternative
5. Hardware verdict
6. Implementation difficulty
7. Initial architecture
8. FYP-I proof-of-concept plan
9. FYP-I prototype plan
10. FYP-II completion plan
```

Output

`Technical Feasibility and Initial Architecture Report`

Stage 14: Scope, Evaluation, Risk, and Delivery Planning

Purpose

Reduce the idea to a realistic and measurable FYP.

Inputs

- Feasibility report
- Service concept
- Novelty audit
- University requirements

Prompt

```
You are acting as an FYP Scope, Evaluation, Risk, and Delivery Specialist.

For each surviving concept, divide the project into:

A. Core problem

The one main stakeholder problem being solved.

B. Core technical contribution

The one main intelligent, algorithmic, optimization, security, data, or systems
contribution.

C. Minimum viable service

The smallest complete end-to-end workflow.

D. Supporting features

Only features required for usability, administration, testing, or evaluation.

E. Optional extensions

Features that may be attempted only after the core project is stable.

F. Explicit exclusions

Features that will deliberately not be developed.
```

```
Limit the recommended project to:

- one primary problem;
- one main stakeholder group;
- one main service workflow;
- one major technical contribution;
- and a manageable number of supporting modules.

For every removed feature, explain why it should be excluded.

Create three scopes:

1. Minimum defensible FYP
2. Recommended balanced FYP
3. Ambitious extension

Then create an evaluation plan.

Include:

1. Evaluation questions
2. Baselines
3. Technical metrics
4. Service metrics
5. User-evaluation method
6. Failure analysis
7. Success thresholds
8. Required participants
9. Required experts
10. Evaluation risks

Then create a risk register.

For each risk, include:

- description;
- probability;
- impact;
- early warning sign;
- mitigation;
- contingency;
- and responsible group member.

Finally create a delivery plan:

Pre-August:
- idea investigation;
```

```
- dataset investigation;
- comparison;
- feasibility;
- defense preparation.

August:
- proposal defense;
- feedback;
- scope revision;
- approval.

FYP-I Iteration 1:
- baseline;
- proof of concept;
- initial data pipeline;
- initial interface;
- one limited workflow.

FYP-I Iteration 2:
- improved technical component;
- module integration;
- prototype;
- initial testing;
- demonstrable workflow.

End of FYP-I:
- two iterations completed;
- prototype demonstrated;
- feasibility established.

FYP-II:
- complete core development;
- integrate all essential modules;
- perform evaluation;
- improve reliability.

By mid-April:
- practical system complete;
- stable job-fair demonstration;
- major evaluation complete.

Mid-April to May:
- formal testing;
- deployment;
- documentation;
- final refinement.
```

```
Reject concepts that cannot be meaningfully evaluated or cannot produce a
demonstrable FYP-I prototype.
```

Output

`Scoped FYP Plan, Evaluation Plan, Risk Register, and Delivery Roadmap`

Stage 15: Final Selection and Proposal Defense Package

Purpose

Select the strongest concept and prepare it for the August defense.

Inputs

- All previous stage outputs

Prompt

```
You are acting as the Final FYP Selection and Proposal Defense Committee.

Review all surviving concepts.

Score each concept using:

1. Problem importance – 10%
2. Evidence strength – 8%
3. Stakeholder clarity – 8%
4. Stakeholder access – 6%
5. Difference from previous FYPs – 12%
6. Difference from contemporary systems – 8%
7. Data feasibility – 10%
8. Technical depth – 10%
9. End-to-end service value – 8%
10. Evaluation strength – 6%
11. Group capability match – 5%
12. FYP-I prototype feasibility – 4%
13. Mid-April completion feasibility – 3%
14. Job-fair demonstration potential – 2%

Apply penalties for:
```

```
- unavailable data;
- inaccessible stakeholders;
- excessive scope;
- weak novelty;
- dependence on paid APIs;
- uncertain institutional approval;
- unsafe outputs;
- weak evaluation;
- inability to complete two FYP-I iterations;
- and inability to finish the practical system by mid-April.

Produce:

1. Full scoring matrix
2. Explanation for every score
3. Strongest reason to select each concept
4. Strongest reason to reject each concept
5. Top three concepts
6. One backup concept
7. Final recommended concept
8. Sensitivity analysis
9. Conditions under which the ranking would change

Then prepare a complete Proposal Defense Package for the recommended concept.

The package must include:

1. Project title
2. One-sentence concept
3. Domain
4. Domain boundaries
5. Background
6. Primary stakeholder
7. Secondary stakeholders
8. Stakeholder-access plan
9. Current workflow
10. Problem statement
11. Evidence that the problem exists
12. Consequences of the problem
13. Contemporary commercial systems
14. Institutional systems
15. Open-source systems
16. Recent research
17. Limitations of existing systems
18. Similar previous university FYPs
19. Comparison with previous FYPs
20. Proposed service
```

```
21. End-to-end workflow
22. Core technical contribution
23. Why the technical component is necessary
24. Required data
25. Preferred dataset
26. Dataset limitations
27. Local-data requirements
28. Fallback data plan
29. Proposed technology stack
30. Initial architecture
31. Core modules
32. Supporting modules
33. Optional modules
34. Explicit exclusions
35. Evaluation questions
36. Baselines
37. Technical metrics
38. Service metrics
39. User-evaluation plan
40. Ethical considerations
41. Privacy considerations
42. Security considerations
43. Hardware requirements
44. Cost estimate
45. FYP-I Iteration 1 plan
46. FYP-I Iteration 2 plan
47. End-of-FYP-I prototype
48. FYP-II plan
49. Mid-April completion plan
50. Job-fair demonstration
51. Testing plan
52. Deployment plan
53. Documentation plan
54. Risk register
55. Risk-mitigation plan
56. Expected outcomes
57. Possible publication angle
58. SDG alignment where appropriate
59. Why the university should approve the idea
60. Why the university might reject the idea
61. Questions the defense panel is likely to ask
62. Suggested answers to likely questions
63. Items that must be validated before the defense

Every factual claim based on online search must include a citation.

Do not present unsupported claims of novelty.
```

Output

`Final Proposal Defense Package`

Phase B: Post-Defense Revision

Stage 16: Defense Feedback, Revision, and Scope Freeze

Purpose

Convert panel feedback into an approved, controlled implementation plan.

Inputs

- Proposal Defense Package
- Defense panel comments
- Supervisor comments

Prompt

```
You are acting as a Post-Defense FYP Revision Committee.

Review:

1. The approved proposal
2. Defense panel comments
3. Supervisor comments
4. Conditions attached to approval
5. Requested changes
6. Rejected features
7. Required comparisons
8. Required dataset changes
9. Required technical changes
10. Required evaluation changes

Classify every comment as:

- Mandatory before implementation
- Mandatory during FYP-I
```

```
- Mandatory during FYP-II
- Recommended
- Optional
- Rejected or inconsistent

For every comment:

1. Explain what it means
2. Identify affected modules
3. Identify affected data
4. Identify affected architecture
5. Identify affected timeline
6. Identify affected evaluation
7. Recommend an action
8. Assign responsibility
9. Assign a deadline

Then produce:

1. Revised problem statement
2. Revised scope
3. Revised architecture
4. Revised dataset plan
5. Revised evaluation plan
6. Revised risk register
7. Revised FYP-I Iteration 1 plan
8. Revised FYP-I Iteration 2 plan
9. Revised FYP-II plan
10. Features removed
11. Features frozen
12. Features deferred
13. Final approved success criteria

The output should create a scope baseline.

No new major feature should be added later without a formal scope-change
decision.
```

Output

`Post-Defense Scope Baseline`

Phase C: FYP-I Development

Stage 17: FYP-I Iteration 1 Planning and Review

Purpose

Build and evaluate the first technical proof of concept.

Inputs

- Scope baseline
- Architecture
- Dataset plan
- Evaluation plan

Prompt

```
You are acting as the FYP-I Iteration 1 Planning and Review Committee.

The objective of Iteration 1 is to prove that the central idea is technically
possible.

Create an Iteration 1 plan containing:

1. Iteration objective
2. Central hypothesis
3. Selected baseline
4. Minimum dataset
5. Data-acquisition tasks
6. Data-cleaning tasks
7. Initial technical method
8. Initial backend tasks
9. Initial frontend tasks
10. Initial database tasks
11. One limited end-to-end workflow
12. Testing tasks
13. Demonstration scenario
14. Responsible group member for every task
15. Dependencies
16. Risks
17. Acceptance criteria
18. Iteration deadline

The Iteration 1 output should include:

- an initial data pipeline;
- a baseline model or algorithm;
```

```
- a proof of concept;
- basic system integration;
- and one demonstrable workflow.

At the end of the iteration, review:

1. What was completed
2. What failed
3. Technical results
4. Data problems
5. Integration problems
6. User-interface problems
7. Performance problems
8. Scope problems
9. New risks
10. Whether the central idea remains feasible

Classify the outcome as:

- Successful
- Successful with revisions
- Major redesign required
- Concept no longer feasible

Do not proceed to Iteration 2 without recording Iteration 1 evidence and
decisions.
```

Output

- `FYP-I Iteration 1 Plan`
- `FYP-I Iteration 1 Review`

Stage 18: FYP-I Iteration 2 Planning and Review

Purpose

Convert the proof of concept into an integrated prototype.

Inputs

- Iteration 1 review
- Revised scope
- Architecture
- Evaluation plan

Prompt

```
You are acting as the FYP-I Iteration 2 Planning and Review Committee.

The objective of Iteration 2 is to produce an integrated and demonstrable
prototype.

Review the results of Iteration 1.

Create an Iteration 2 plan containing:

1. Problems that must be corrected
2. Technical component improvements
3. Dataset improvements
4. Model or algorithm improvements
5. Backend integration
6. Frontend integration
7. Database integration
8. User-role implementation
9. Core workflow implementation
10. Logging
11. Error handling
12. Basic security
13. Initial usability improvements
14. Initial technical testing
15. Initial user or stakeholder feedback
16. Demonstration scenario
17. Acceptance criteria
18. Responsible group member for each task
19. Dependencies
20. Risks
21. Iteration deadline

The prototype should demonstrate:

1. User or system input
2. Input validation
3. Core technical processing
4. Meaningful output
5. User or professional action
6. Record storage or history
7. One complete service workflow

At the end of Iteration 2, review:

1. Completed modules
```

```
2. Incomplete modules
3. Technical performance
4. Integration quality
5. Prototype usability
6. Data quality
7. Evaluation readiness
8. Risks transferred to FYP-II
9. Scope changes
10. Whether the prototype is suitable for the FYP-I demonstration

Classify the result as:

- Strong FYP-I prototype
- Acceptable prototype with limitations
- Weak prototype requiring urgent revision
- Prototype does not validate the proposal
```

Output

- `FYP-I Iteration 2 Plan`
- `FYP-I Iteration 2 Review`

Stage 19: End-of-FYP-I Prototype Assessment

Purpose

Assess whether FYP-I has genuinely established project viability.

Inputs

- Iteration 1 review
- Iteration 2 review
- Prototype
- Initial results

Prompt

```
You are acting as the End-of-FYP-I Prototype Evaluation Panel.

Evaluate whether the group has completed two meaningful iterations and produced
a valid prototype.

Assess:
```

```
1. Problem understanding
2. Stakeholder relevance
3. Dataset readiness
4. Technical feasibility
5. Baseline performance
6. Core technical contribution
7. Architecture implementation
8. Module integration
9. End-to-end workflow
10. Interface usability
11. Demonstration stability
12. Error handling
13. Initial evaluation
14. Documentation
15. Code quality
16. Risk management
17. FYP-II readiness

Determine whether the prototype is:

- an integrated service;
- a partial service;
- a disconnected collection of components;
- or only a technical demonstration.

Produce:

1. Completed FYP-I deliverables
2. Missing FYP-I deliverables
3. Technical debt
4. Data debt
5. Integration debt
6. Evaluation debt
7. Documentation debt
8. FYP-II priorities
9. Features to remove
10. Features to retain
11. Features to defer
12. Revised mid-April completion plan
13. Final FYP-I verdict
```

Output

`End-of-FYP-I Prototype Assessment`

Phase D: FYP-II Completion, Job Fair, Testing, Deployment, and Documentation

---

Stage 20: FYP-II Completion Planning

Purpose

Turn the FYP-I prototype into a complete practical system by mid-April.

Inputs

- FYP-I assessment
- Revised scope
- Evaluation plan
- Risk register

Prompt

```
You are acting as an FYP-II Project Completion Planner.

The practical system must be substantially complete by mid-April.

Use the FYP-I assessment to create an FYP-II plan.

Organize remaining work into:

A. Core development

1. Remaining technical features
2. Remaining service modules
3. Required integrations
4. Data improvements
5. Model or algorithm improvements
6. Backend completion
7. Frontend completion
8. Database completion
9. Security improvements
10. Error handling

B. Evaluation

1. Technical evaluation
```

```
2. Baseline comparison
3. Service evaluation
4. User testing
5. Failure analysis
6. Performance analysis

C. Reliability

1. Regression testing
2. Stress testing
3. Edge cases
4. Recovery procedures
5. Logging
6. Monitoring

D. Job-fair preparation

1. Demonstration mode
2. Demo data
3. Backup demonstration
4. Poster
5. Presentation
6. Booth setup
7. Explanation of technical contribution
8. Explanation of results

Separate all remaining work into:

- Must complete before mid-April
- Should complete before mid-April
- May complete after the job fair
- Must be removed from scope

Create:

1. Work breakdown structure
2. Responsibility matrix
3. Weekly schedule
4. Critical path
5. Risk-adjusted plan
6. Scope-drop sequence
7. Mid-April completion criteria
8. Job-fair readiness criteria
```

Output

`FYP-II Practical Completion Plan`

Stage 21: Technical, Service, and User Evaluation

Purpose

Produce defensible evidence that the system works and improves the target workflow.

Inputs

- Evaluation plan
- Working system
- Datasets
- Baselines
- Users or stakeholders

Prompt

```
You are acting as an FYP Evaluation and Experimentation Committee.

Design and manage the complete evaluation of the system.

A. Evaluation questions

Define the exact questions the evaluation must answer.

B. Baselines

Identify:

- manual workflow;
- simple heuristic;
- existing model;
- existing system;
- keyword search;
- non-personalized version;
- or other appropriate baseline.

C. Technical metrics

Select only relevant metrics, such as:

- accuracy;
- precision;
- recall;
- F1-score;
```

```
- mean absolute error;
- latency;
- throughput;
- retrieval relevance;
- groundedness;
- hallucination rate;
- false-alert rate;
- recommendation acceptance;
- route cost;
- resource consumption;
- robustness;
- or task success.

D. Service metrics

Measure:

- time saved;
- steps reduced;
- workload reduced;
- errors reduced;
- response speed;
- task completion;
- resolution rate;
- or cost improvement.

E. User evaluation

Define:

1. Participant group
2. Participant count
3. Recruitment method
4. Tasks
5. Questionnaire
6. Interviews
7. Usability measures
8. Comparison procedure
9. Consent process
10. Privacy protection

F. Failure analysis

Categorize:

- false positives;
- false negatives;
- hallucinations;
```

```
- retrieval failures;
- interface failures;
- integration failures;
- and user misunderstandings.

G. Reporting

Produce:

1. Experiment design
2. Dataset split
3. Baseline results
4. Proposed-system results
5. Statistical comparison where appropriate
6. Qualitative findings
7. Failure cases
8. Limitations
9. Threats to validity
10. Final evaluation conclusion

Major evaluation should be completed before the mid-April job fair.

Only supplementary testing should remain afterward.
```

Output

`FYP Evaluation Report`

Stage 22: Mid-April Job-Fair Readiness

Purpose

Prepare a stable and professional demonstration for visiting companies.

Inputs

- Working system
- Evaluation results
- Architecture
- Project summary

Prompt

```
You are acting as a University Job-Fair FYP Demonstration Adviser.

Prepare the project for repeated demonstration to visiting companies.

Create a job-fair readiness plan containing:

1. Thirty-second project explanation
2. Two-minute project explanation
3. Five-minute demonstration script
4. Main stakeholder problem
5. Existing-system limitation
6. Proposed service
7. Technical contribution
8. Data used
9. Main result
10. Practical impact
11. Architecture explanation
12. Limitations
13. Future work

Define the primary demonstration workflow.

Also prepare:

- backup demonstration;
- recorded demonstration;
- offline mode where possible;
- local dataset;
- test accounts;
- reset procedure;
- failure-recovery procedure;
- troubleshooting checklist;
- booth poster content;
- architecture diagram content;
- results chart content;
- QR code or repository plan where allowed;
- and frequently asked questions.

Test job-fair readiness against:

1. Internet failure
2. API failure
3. Cloud failure
4. Model timeout
```

```
5. Database failure
6. Invalid input
7. Repeated demonstrations
8. Different screen sizes
9. Unexpected stakeholder questions
10. Technical questions from company representatives

Produce:

1. Job-fair checklist
2. Demonstration script
3. Backup plan
4. Booth content
5. Technical-question bank
6. Non-technical-question bank
7. Final readiness verdict
```

Output

`Job-Fair Demonstration Package`

Stage 23: Post-Job-Fair Testing, Deployment, and Documentation

Purpose

Complete the formal work remaining between mid-April and May.

Inputs

- Job-fair feedback
- Working system
- Evaluation report
- University documentation requirements

Prompt

```
You are acting as the Final FYP Testing, Deployment, and Documentation
Committee.

The practical system has already been completed and demonstrated.
```

Use the remaining period to strengthen:

A. Formal testing

1. Functional testing
2. Integration testing
3. Regression testing
4. Performance testing
5. Security testing
6. Usability testing
7. Failure-case testing
8. Compatibility testing
9. Data-validation testing
10. Deployment testing

B. Deployment

1. Deployment environment
2. Server configuration
3. Database deployment
4. Model deployment
5. API configuration
6. Secret management
7. Logging
8. Monitoring
9. Backup
10. Recovery
11. Installation instructions
12. Environment reproduction

C. Documentation

1. Final report
2. Problem background
3. Literature and contemporary systems
4. Requirements
5. Architecture
6. Methodology
7. Data
8. Implementation
9. Experiments
10. Results
11. Testing
12. Deployment
13. Security
14. Ethics
15. Limitations
16. Future work

```
17. User manual
18. Administrator manual
19. Installation guide
20. API documentation
21. Maintenance notes

D. Final refinement

1. Job-fair feedback
2. Remaining defects
3. Interface improvements
4. Deployment defects
5. Reliability improvements
6. Final presentation
7. Final viva preparation

Produce:

1. Final testing plan
2. Test-case catalogue
3. Deployment checklist
4. Deployment architecture
5. Documentation outline
6. Final report checklist
7. Final presentation structure
8. Final viva questions
9. Submission checklist
10. Final completion verdict
```

Output

`Final Testing, Deployment, and Documentation Package`

---

7. Master Controller Prompt

Use the following prompt to tell the AI how the overall workflow should be managed.

Students should still execute each stage separately.

```
You are coordinating a multi-stage Final Year Project workflow.

The student group has already been formed.
```

```
The FYP lifecycle is:

Phase A:
Pre-August idea discovery, validation, feasibility analysis, and proposal
preparation

Phase B:
August proposal defense, panel feedback, revision, and scope freeze

Phase C:
FYP-I with two required development iterations and an end-of-semester
demonstrable prototype

Phase D:
FYP-II practical completion by mid-April

Phase E:
Mid-April university job fair

Phase F:
Mid-April to May formal testing, deployment, documentation, and final refinement

You have access to:

1. A repository of previous university FYPs
2. The student group profile
3. University and supervisor requirements
4. Online search
5. Contemporary products
6. Open-source systems
7. Research papers
8. Datasets
9. Technical documentation

Execute the stages in order:

1. Repository Intelligence
2. Group Capability and Constraint Profile
3. Domain Exploration
4. Problem-Space Discovery
5. Stakeholder and Workflow Validation
6. Problem Evidence Validation
7. Contemporary Systems and Research Investigation
8. Dataset and Data-Source Investigation
9. Opportunity Ranking
10. Solution-Direction Generation
11. Service-System Design
12. Novelty and Duplication Audit
```

```
13. Technical Feasibility and Initial Architecture
14. Scope, Evaluation, Risk, and Delivery Planning
15. Final Selection and Proposal Defense Package
16. Defense Feedback, Revision, and Scope Freeze
17. FYP-I Iteration 1
18. FYP-I Iteration 2
19. End-of-FYP-I Prototype Assessment
20. FYP-II Completion Planning
21. Technical, Service, and User Evaluation
22. Mid-April Job-Fair Readiness
23. Post-Job-Fair Testing, Deployment, and Documentation

At the beginning of every stage:

- state the objective;
- list the inputs;
- identify missing information;
- state the decision criteria;
- and identify the relevant deadline.

At the end of every stage:

- provide structured outputs;
- list retained items;
- list rejected items;
- explain all rejections;
- list unresolved questions;
- identify assumptions;
- identify risks;
- state whether the workflow should proceed;
- and prepare a handoff for the next stage.

Use online search to verify:

- problem evidence;
- stakeholder demand;
- contemporary systems;
- commercial products;
- research;
- open-source alternatives;
- datasets;
- licenses;
- APIs;
- hardware requirements;
- costs;
- regulations;
- evaluation methods;
- and deployment feasibility.
```

```
Use the previous FYP repository to detect duplication.

Do not allow an idea to proceed merely because:

- it sounds modern;
- it uses AI;
- it uses an LLM;
- it uses RAG;
- it uses agents;
- it contains many features;
- or the exact title does not appear in the repository.

Generator stages should be creative.

Evaluation stages should be skeptical.

Feasibility stages should be conservative.

Scope stages should remove unnecessary features.

The final selection should be based on evidence rather than enthusiasm.

The project must:

- survive the August proposal defense;
- complete two meaningful FYP-I iterations;
- produce a demonstrable FYP-I prototype;
- reach practical completion by mid-April;
- and complete formal testing, deployment, and documentation by May.
```

---

8. Standard Handoff Template

At the end of every stage, ask the AI to use this structure:

```
STAGE COMPLETED:
[Stage name]

FYP PHASE:
[Pre-defense / Post-defense / FYP-I / FYP-II / Job fair / Finalization]

DEADLINE STATUS:
[Is the work still aligned with the August defense, FYP-I prototype, mid-April
completion, and May finalization?]
```

```
INPUTS USED:
[List]

OUTPUTS CREATED:
[List]

ITEMS RETAINED:
[List]

ITEMS REJECTED:
[List with reasons]

DECISIONS:
[List]

UNRESOLVED QUESTIONS:
[List]

ASSUMPTIONS:
[List]

WEB SOURCES USED:
[List with publication dates]

RISKS IDENTIFIED:
[List]

SCOPE CHANGES:
[List]

DECISION:
- Proceed
- Repeat this stage
- Return to an earlier stage
- Redesign concept
- Reduce scope
- Stop workflow

NEXT-STAGE HANDOFF:
[Structured summary]
```

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9. Proposal-Defense Quality Gates

An idea should not be defended unless it achieves at least:

Requirement	Minimum score
Problem evidence	7/10
Stakeholder clarity	8/10
Workflow understanding	8/10
Stakeholder access	6/10
Contemporary-system analysis	7/10
Previous-FYP differentiation	7/10
Dataset feasibility	7/10
Technical feasibility	7/10
Service completeness	7/10
Evaluation strength	7/10
Group capability match	7/10
FYP-I prototype feasibility	8/10
Mid-April completion feasibility	8/10

Critical categories are:

- stakeholder clarity;
- problem evidence;
- dataset feasibility;
- technical feasibility;
- evaluation feasibility;
- FYP-I prototype feasibility;
- and mid-April completion feasibility.

A high total score should not compensate for failure in a critical category.

---

10. FYP-I Quality Gates

The group should not complete FYP-I without:

- two recorded development iterations;
- an initial dataset or data pipeline;
- a baseline technical method;
- an improved technical method or integrated component;
- a functional backend;
- a usable interface;

- a database or persistent record system where relevant;
- one complete end-to-end workflow;
- initial testing;
- and a demonstrable prototype.

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11. FYP-II Mid-April Quality Gates

By mid-April, the project should have:

- all core features completed;
- all essential modules integrated;
- a stable main workflow;
- major technical evaluation completed;
- major user or service evaluation completed;
- acceptable reliability;
- error handling;
- a clear demonstration;
- a backup demonstration;
- and professional job-fair materials.

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12. When to Return to an Earlier Stage

Return to problem discovery when:

- the problem lacks evidence;
- stakeholders cannot be identified;
- users cannot be reached;
- or the problem is too broad.

Return to contemporary-system investigation when:

- a similar product is discovered;
- a similar FYP is found;
- or the novelty claim becomes weak.

Return to dataset investigation when:

- the preferred dataset is unavailable;
- licensing is unsuitable;
- local data are required;
- or data quality is poor.

Return to solution generation when:

- the problem is strong;
- but the solution is only a chatbot, classifier, or dashboard.

Return to service design when:

- the technical component is strong;
- but there is no end-to-end service.

Return to scope reduction when:

- too many features are added;
- FYP-I cannot produce a prototype;
- or mid-April completion becomes unlikely.

Reject the concept when:

- data cannot be obtained;
- the main stakeholder cannot be reached or understood;
- the problem cannot be measured;
- the project substantially duplicates an existing system;
- the core contribution is too weak;
- or the project cannot be completed within the FYP lifecycle.

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13. Final Advice to Students

Do not select an FYP idea only because:

- it uses a popular technology;
- it has an impressive title;
- an AI system describes it as innovative;
- it contains many modules;
- it resembles a startup;
- or it can be demonstrated through a chatbot.

A strong FYP must survive six tests.

1. Problem test

Does a real stakeholder experience a real and meaningful problem?

2. Evidence test

Can the group prove that the problem exists?

3. Difference test

Is the proposed contribution meaningfully different from previous FYPs and contemporary systems?

4. Data test

Can the required data be obtained, used, and prepared?

5. Feasibility test

Can the group complete two FYP-I iterations, produce a prototype, and finish the practical system by mid-April?

6. Evaluation test

Can the group measure whether the project works and improves the target workflow?

Students should prefer:

> A smaller, complete, integrated, and well-evaluated project

over:

> A large platform containing many unfinished features.

The final project should not merely demonstrate technology.

It should provide a meaningful service to a clearly defined stakeholder and produce evidence that the service improves an existing workflow.